

Recorded clean-source, hosted, and fresh-clone checks

Repository source: CLEAN_CLONE_CHECK.md

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• Record date: September 19, 2026	
• Current package version: v1.0.0	
• Audited pre-release package: v1.0.0-rc1	
• Canonical repository: https://github.com/swihart/minvol-degree3-contact-bound	
• Visibility during this audit: private	
• Audited candidate commit: 892d4bc6fad07950e78da66e45b95063a6415af6	
• Hosted workflow: https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663	

This document preserves the clean-source, hosted continuous-integration, and fresh-clone evidence for the audited pre-release state. The final v1.0.0 tag uses the same replay requirements and must pass its own hosted and tagged-clone gates before the GitHub Release is published.

Results at a glance

Checkpoint	Result
Standalone certificate (21a1750)	Exact Python, binary64, ten tests, base R, checksums, and clean status passed
Adversarial mutation suite (068be1f)	Seven representative corruptions rejected
Focused Paper v1 (b11a13a)	Paper reconciliation, builds, paired documents, and PDF checks passed
Complete release documentation (dc4017a)	Release-document, exact-constants, link, checksum, and PDF checks passed
Release-candidate metadata (892d4bc)	Author, citation, repository, version, and split-license metadata passed
Hosted candidate workflow	Three of three jobs completed successfully on 892d4bc
Remote fresh-clone replay	Full Python/R replay, paper build, 22 document pairs, complete bundle, and clean status passed

1. Curated fresh history

The audited bundle contains exactly five fresh-history commits. The labels below identify their public audit roles; the hashes are authoritative:

21a175010fb18d74e11286ef0e1135cb32c47053	Standalone exact-certificate checkpoint
068be1ff4bd7d815714445ec24549715eef0e5aa	Adversarial mutation-test checkpoint
b11a13a145ebb09218dca542057b1d44700cb48e	Focused Paper v1 checkpoint
dc4017ae564f8680eddbb9f1b4407ff1bd45991e	Complete release-document checkpoint

892d4bc6fad07950e78da66e45b95063a6415af6 Release-candidate metadata checkpoint

The literal historical subject of commit `b11a13a` used lower-case *lean* in the ordinary sense of *concise*. Public documentation uses **focused** to avoid confusion with the Lean theorem prover. No proof-assistant formalization is claimed.

`git fsck --full` passed, the bundle records a complete SHA-1 history, and no private research branch or private repository history is present.

2. Transfer integrity

The four audit-transfer files were checked against the uploaded manifest. The manifest itself has SHA-256

3f958520e0e119cad4db97d60a2ac3702b8144a8f6228068bdfef6d43124f57d

The three transferred evidence artifacts are:

Main-branch bundle

```
filename: minvol-degree3-contact-bound-v1.0.0-rc1-main.bundle
SHA-256: 2c324230f02ba4cb2541e0d353cc7906ae460cac0a768267144f650a0278a1a3
```

Fresh-clone replay transcript

```
filename: minvol-degree3-contact-bound-v1.0.0-rc1-fresh-replay.txt
SHA-256: ce2abd06646fdf89d6e60b9069ee55451715a35970afedd24e4acf0d1d64de7f
```

GitHub Actions log archive

```
filename: minvol-degree3-contact-bound-v1.0.0-rc1-github-actions-logs.zip
SHA-256: 44fc68b36e05153fabd2a93aaa4b092b946016b9eb10bcecf37d7bf5c6b4ad3a
```

The GitHub Actions ZIP contains 36 entries, passes ZIP integrity testing, and expands to 564,330 bytes. These raw transfer artifacts are audit inputs rather than tracked theorem objects. Their hashes and conclusions are indexed in `certificate/evidence/rc1_remote_audit.json`.

3. Hosted workflow on the audited candidate

Workflow run <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663> checked out commit `892d4bc6fad07950e78da66e45b95063a6415af6` in all three jobs. The named author reported all three jobs green, and the downloaded logs contain all required pass markers with no error marker or nonzero command exit.

Hosted job	Principal evidence	Result
Exact certificate, audits, metadata, release docs, and paper reconciliation	Exact verifier; binary64 audit; 10 regression tests; 7 mutation tests; metadata, release-doc, and paper checks	PASS
Independent base-R audit	Base R reconstruction on R 4.6.1	PASS
Build and preflight paper and documentation PDFs	Ten-page paper; 22 Markdown/PDF pairs; committed and rebuilt Poppler preflight	PASS

The hosted runners used Ubuntu 24.04.5 LTS, runner image `ubuntu-24.04` version `20260907.300.1`. The Python job used CPython 3.13.15 and NumPy 2.3.5. The R job used R 4.6.1. The document job used Pandoc 3.10.1, TeX Live 2023/Debian, and Poppler.

The document job also uploaded a temporary typeset-PDF artifact. Its recorded run-scoped URL is <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663/artifacts/10591743752>. That artifact is not an immutable release asset and may expire.

4. Fresh clone from the staging remote

The named author cloned the private staging remote and began the audit at 2026-09-19 16:58:42 EDT. The fresh clone reported:

```
branch: main
HEAD: 892d4bc6fad07950e78da66e45b95063a6415af6
origin/main: 892d4bc6fad07950e78da66e45b95063a6415af6
public-history commits: 5
Python: 3.14.7
NumPy: 2.3.5
R: 4.6.1
TeX Live: 2023
```

Before execution, the root, certificate, paper, and rendered-document checksum manifests passed. The fresh-clone transcript then records all of the following:

```
MINVOL UNIVERSAL CONTACT-BOUND CERTIFICATE: EXACT
MINVOL UNIVERSAL CONTACT-BOUND BINARY64 AUDIT: PASS
Ran 10 tests
OK
Ran 7 tests
OK
MINVOL ADVERSARIAL MUTATION SUITE: PASS
MINVOL EXACT-CONSTANTS APPENDIX: CURRENT
MINVOL RELEASE-DOCUMENT CHECK: PASS
MINVOL CITATION METADATA CHECK: PASS
MINVOL METADATA PREFLIGHT: PASS
MINVOL PAPER-CERTIFICATE RECONCILIATION: PASS
MINVOL UNIVERSAL CONTACT-BOUND BASE-R AUDIT: PASS
MINVOL UNIVERSAL CONTACT-BOUND REPLAY: PASS
MINVOL PAPER BUILD: PASS
MINVOL MARKDOWN PDF BUILD: PASS
MINVOL MARKDOWN/PDF PAIR CHECK: PASS
```

The paper rebuilt to ten pages. Both the 22 committed Markdown/PDF pairs and the 22 local rebuild pairs passed. The Mac did not run Poppler locally; the successful hosted document job supplies that required page/font/text preflight.

After all replay and build steps, `git status --short` was empty. The bundle created from that fresh clone points `refs/heads/main` to the audited candidate and records complete history.

5. Interpretation

This evidence shows that the audited pre-release state is self-contained, replays from the actual remote on the named author's machine, and passes the three-job hosted workflow. It also confirms that the public-format history is the intended five-commit history and that the replay leaves tracked source unchanged.

It does **not** establish external mathematical reproduction, peer review, novelty, correctness of every conceptual reduction, sharpness, a minimizing body, or Meissner extremality. The fresh-clone audit was performed by the named author, and the hosted jobs execute project-supplied programs.

6. Tagged-release follow-up

The historical audit above is complete and intentionally retains its `rc1` filenames and commit identifiers. Publication of `v1.0.0` additionally requires:

1. a green three-job hosted run on the final release commit;
 2. an annotated `v1.0.0` tag pointing to that exact commit;
 3. a green three-job workflow triggered by the immutable tag;
 4. a complete replay from a fresh checkout of the tag with clean final status;
 5. deterministic release assets built from the tagged checkout and verified against `SHA256SUMS.txt` after download;
- and

6. explicit named-author approval before changing visibility, publishing the GitHub Release, or announcing the result.